

ring with N corners. (See Figure 6.3.) There are only $(N - 1)!$ different diagrams here because a permutation of labels that moves each label to the next vertex around the ring yields the same diagram. Hence such a graph is accompanied with a factor

$$\frac{(N - 1)!}{N!} = \frac{1}{N}. \quad (6.1.17)$$

(vi) In theories involving fermion fields, the use of Eqs. (6.1.4)–(6.1.6) to move annihilation and creation operators to the right and left introduces minus signs into the contribution of various pairings. To be specific, we get a minus sign wherever the permutation of the operators in Eq. (6.1.1) that is required to put all paired operators adjacent to one another (with annihilation operators just to the left of the paired creation operators) involves an odd number of interchanges of fermion operators. (This is because to compute the contribution of a certain pairing, we can first permute all operators in Eq. (6.1.1) so that each annihilation operator is just to the left of the creation operator with which it is paired, ignoring all commutators and anticommutators of unpaired operators, and then replace each product of paired operators with their commutators or anticommutators.) One immediate consequence is to produce the minus sign in the relative sign of the two terms in Eq. (6.1.14) for the fermion propagator. Whatever permutation puts the annihilation part $\psi^+(x)$ of a field in $\mathcal{H}(x)$ just to the left of the creation part $\psi^{+\dagger}(y)$ of a field adjoint in $\mathcal{H}(y)$, the permutation that puts the annihilation part $\psi^{-\dagger}(y)$ of the field adjoint just to the left of the creation part $\psi^-(x)$ of the field involves one extra interchange of fermion operators, yielding the minus sign in the second term of Eq. (6.1.14) for fermions.

In addition, minus signs can rise in the contribution of whole Feynman diagrams. As an example, let us take up a theory in which the sole interaction of fermions takes the form

$$\mathcal{H}(x) = \sum_{\ell mk} g_{\ell mk} \psi_{\ell}^{\dagger}(x) \psi_m(x) \phi_k(x), \quad (6.1.18)$$

where $g_{\ell mk}$ are general constants, $\psi_{\ell}(x)$ are a set of complex fermion fields, and $\phi_m(x)$ are a set of real bosonic (but not necessarily scalar) fields. (Not only quantum electrodynamics, but the whole ‘standard model’ of weak, electromagnetic, and strong interactions, has fermionic interactions that can all be put in this form.) Let us first take up the process of fermion–fermion scattering, $12 \rightarrow 1'2'$, to second order in $\mathcal{H}$. The fermion operators in the second-order term in Eq. (6.1.1) appear in the order (with obvious abbreviations)

$$a(2')a(1')\psi^{\dagger}(x)\psi(x)\psi^{\dagger}(y)\psi(y)a^{\dagger}(1)a^{\dagger}(2). \quad (6.1.19)$$

There are two connected diagrams to this order, corresponding to the